



HELLENIC MEDITERRANEAN UNIVERSITY

SCHOOL OF ENGINEERING

DEPARTMENT OF ELECTRICAL &
COMPUTER ENGINEERING

PROJECT ON

**KNOWLEDGE REPRESENTATION
ON THE WORLD WIDE WEB**

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Primary Objective:

The implementation of this project involves the integration of artificial intelligence tools, such as ChatGPT, to improve the representation of information in a knowledge graph. The primary objective is to process the 50 most frequent or interesting SPARQL queries from the provided DBpedia query log. These queries must be answered using a web API and then articulated in plain English using ChatGPT. The goal is to translate the complex query results into easily understandable English sentences or paragraphs. This translation will help make the information accessible to a broader audience that may not be familiar with technical query languages or complex data structures. The project emphasizes the use of Python and OpenAI tools to efficiently process and represent these queries and responses.

Detailed Steps for Project Implementation:

This project was implemented using Python, which facilitated “communication” with OpenAI’s services (ChatGPT). More specifically, the steps we followed in the corresponding code script we mentioned for this communication are as follows:

Integration with OpenAI and SPARQL Queries: First, we import the necessary libraries into the script, such as `openai`, `urllib`, and `json`, thereby indicating that the code interacts with the OpenAI API and processes web requests and JSON data. This aligns with the project’s goal of using artificial intelligence to interpret the results of SPARQL queries.

API Key: The code includes a section for initializing the OpenAI client with the API key. This configuration is essential for enabling communication between the Python script and the OpenAI services, thereby allowing our code to leverage the capabilities of ChatGPT.

SPARQL Query Execution Function: A significant portion of the code is dedicated to defining the `run_sparql_query` function. This function takes a SPARQL query as input, constructs a URL to query the DBpedia endpoint, and retrieves the results in JSON format. It also demonstrates the process of

converting complex SPARQL queries into a format that can be further processed.

Data retrieval and processing: Next, the code retrieves data from the DBpedia endpoint, a step that is necessary to obtain the data required for its subsequent interpretation and representation. Finally, the results retrieved from the DBpedia endpoint are displayed in the Python terminal. These results include the potential question generated by ChatGPT for each SPARQL query, followed by the corresponding answer.

Below are detailed tables showing the corresponding SPARQL queries and ChatGPT's responses.

SPARQL Query Results Tables

The query submitted to ChatGPT for each different SPARQL query is as follows:

“ Based on this sparql query, *{query}*, create a possible question as if you were asking a human. Do not refer at all at the DBpedia database. After you create the question, please proceed to create an answer for it using info from the sparql result.”

- In the {query} field, the corresponding SPARQL query is entered each time for each query.

Therefore, to save space (and time), we will not list each query separately but only the SPARQL query that corresponds to it.

1)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>PREFIX dbr:
<http://dbpedia.org/resource/>SELECT ?winnerName WHERE {?winner
dbo:award dbr:Nobel_Prize_in_Literature ;foaf:name ?winnerName .} LIMIT 10
```

ChatGPTs' Answer:

Question: Can you provide me with the names of the winners of the Nobel Prize in Literature?

Answer: Sure! Here are the names of some winners of the Nobel Prize in Literature:

1. Ernest Hemingway
2. Gabriel Garcia Marquez
3. Toni Morrison
4. Alice Munro
5. Orhan Pamuk
6. Doris Lessing
7. J.M. Coetzee
8. Mario Vargas Llosa
9. Mo Yan
10. Alice Munro

2)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>PREFIX dbr:
<http://dbpedia.org/resource/>PREFIX dct: <http://purl.org/dc/terms/>SELECT
?battleName WHERE {?battle dct:subject
dbr:Category:Battles_and_operations_of_World_War_II ;foaf:name ?battleName .}
LIMIT 10
```

ChatGPTs' Answer:

Question: Can you provide me with the names of 10 battles and operations that happened during World War II?

Answer: Yes, here are the names of 10 battles and operations that happened during World War II: Battle_of_France, Battle_of_Britain, Operation_Barbarossa, Second_Battle_of_El_Alamein, Battle_of_Stalingrad, Battle_of_Midway, Battle_of_the_Bulge, Operation_Avalanche, Battle_of_Iwo_Jima, Battle_of_Okinawa

3)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>PREFIX dct:
<http://purl.org/dc/terms/>SELECT ?musicianName WHERE {?musician
dct:subject <http://dbpedia.org/resource/Category:American_jazz_musicians>
;foaf:name ?musicianName .} LIMIT 10
```

ChatGPTs' Answer:

Question: Can you give me the names of 10 American jazz musicians?

Answer: Sure! Here are the names of 10 American jazz musicians:

1. John Coltrane
2. Duke Ellington
3. Louis Armstrong
4. Miles Davis
5. Thelonious Monk
6. Charlie Parker
7. Billie Holiday
8. Ella Fitzgerald
9. Dizzy Gillespie
10. Herbie Hancock

4)

Corresponding SPARQL Code:

```
PREFIX      dbo:      <http://dbpedia.org/ontology/>PREFIX      dbr:
<http://dbpedia.org/resource/>PREFIX  dct:  <http://purl.org/dc/terms/>SELECT
?siteName      WHERE      {?site      dct:subject
dbr:Category:World_Heritage_Sites_in_France ;foaf:name ?siteName .} ORDER
BY RAND() LIMIT 10
```

ChatGPTs' Answer:

Question: Can you provide me with the names of 10 random World Heritage Sites in France?

Answer: Some of the random World Heritage Sites in France are:

1. Mont-Saint-Michel and its Bay
2. Palace and Park of Versailles
3. Vézelay, Church and Hill
4. Decorated Grottoes of the Vézère Valley
5. Prehistoric Sites and Decorated Caves of the Vézère Valley
6. Amiens Cathedral
7. Cistercian Abbey of Fontenay
8. Palace and Park of Fontainebleau
9. Roman Theatre and its Surroundings and the "Triumphal Arch" of Orange
10. From the Great Saltworks of Salins-les-Bains to the Royal Saltworks of Arc-et-Senans, the Production of Open-pan Salt

5)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>PREFIX dbr:
<http://dbpedia.org/resource/>SELECT ?cityName WHERE {?city a dbo:City
;dbo:country dbr:Japan ;foaf:name ?cityName .} ORDER BY RAND() LIMIT 15
```

ChatGPTs' Answer:

Question: Can you provide me with 15 names of cities in Japan?

Answer: Sure! Here are 15 random city names in Japan: Nagoya, Hiroshima, Fukuoka, Kawasaki, Sapporo, Sendai, Okayama, Saitama, Chiba, Shizuoka, Yokohama, Kyoto, Osaka, Kobe, and Tokyo.

6)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>PREFIX dbr:
<http://dbpedia.org/resource/>PREFIX dct: <http://purl.org/dc/terms/>SELECT
?dishName WHERE {?dish dct:subject dbr:Category:Italian_cuisine ;foaf:name
?dishName .} LIMIT 20
```

ChatGPTs' Answer:

Question: Can you provide some names of Italian dishes?

Answer: Sure! Here are some names of Italian dishes: Ragù alla bolognese, Margherita pizza, Lasagna, Cannoli, Tiramisu, Carbonara, Caprese salad, Prosciutto e melone, Pesto pasta, Gelato, Bistecca alla fiorentina, Arancini, Osso buco, Gnocchi, Panettone, Bruschetta, Minestrone, Spaghetti aglio e olio, Risotto, and Focaccia.

7)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>PREFIX dbr:
<http://dbpedia.org/resource/>SELECT ?workTitle WHERE {?work dbo:author
dbr:William_Shakespeare ;foaf:name ?workTitle .} LIMIT 15
```

ChatGPTs' Answer:

Question: What are some works written by William Shakespeare?

Answer: Some works written by William Shakespeare include "Hamlet", "Macbeth", "Romeo and Juliet", "Othello", "A Midsummer Night's Dream", "The Tempest", "Julius Caesar", "King Lear", "Much Ado About Nothing", "Twelfth Night", "As You Like It", "The Merchant of Venice", "Antony and Cleopatra", "Richard III", and "The Taming of the Shrew".

8)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>SELECT ?islandName WHERE
{?island a dbo:Island ;dbo:location <http://dbpedia.org/resource/Caribbean>
;foaf:name ?islandName .} LIMIT 15
```

ChatGPTs' Answer:

Question: Can you give me the names of some islands located in the Caribbean?

Answer: Sure! Some of the names of the islands located in the Caribbean are:

1. Cuba
2. Hispaniola
3. Jamaica
4. Puerto Rico
5. The Bahamas
6. Grenada
7. Trinidad and Tobago
8. Barbados
9. Dominica
10. Saint Lucia
11. Saint Kitts and Nevis
12. Antigua and Barbuda
13. Saint Vincent and the Grenadines
14. Aruba
15. Curacao

9)

Corresponding SPARQL Code:

```
PREFIX owl: <http://www.w3.org/2002/07/owl#> PREFIX xsd:
<http://www.w3.org/2001/XMLSchema#> PREFIX rdfs:
<http://www.w3.org/2000/01/rdf-schema#> PREFIX rdf:
<http://www.w3.org/1999/02/22-rdf-syntax-ns#> PREFIX foaf:
<http://xmlns.com/foaf/0.1/> PREFIX dc: <http://purl.org/dc/elements/1.1/> PREFIX
: <http://dbpedia.org/resource/> PREFIX dbpedia2: <http://dbpedia.org/property/>
PREFIX dbpedia: <http://dbpedia.org/> PREFIX skos:
<http://www.w3.org/2004/02/skos/core#> SELECT ?name from <http://dbpedia3.8>
where { ?drink dct:subject
<http://dbpedia.org/resource/Category:Cocktails_with_whisky> ; rdfs:label ?name .
FILTER ( lang(?name) = "en" ) }
```

ChatGPTs' Answer:

Question: Can you provide some names of cocktails with whisky from the DBpedia database?

Answer: Based on the SPARQL query result, one possible cocktail with whisky is "Whisky Sour".

10)

Corresponding SPARQL Code:

```
select ?actor from <http://dbpedia3.8> where {
<http://dbpedia.org/resource/Monsieur_Klein><http://dbpedia.org/ontology/starring
> ?actor. }
```


ChatGPTs' Answer:

Question: Who are the actors that starred in the film "Monsieur Klein"?

Answer: The actors that starred in the film "Monsieur Klein" are Jeanne Moreau, Jean Rochefort, and Jean-Pierre Marielle.

11)

Corresponding SPARQL Code:

```
select ?actor from <http://dbpedia3.8> where {  
<http://dbpedia.org/resource/The_Bostonians_(film)><http://dbpedia.org/ontology/s  
tarring> ?actor. }
```

ChatGPTs' Answer:

Question: Who starred in the film "The Bostonians"?

Answer: The actors who starred in the film "The Bostonians" are Vanessa Redgrave, Christopher Reeve, and Jessica Tandy, among others.

12)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/> PREFIX dbr:  
<http://dbpedia.org/resource/>SELECT ?winnerName WHERE {?winner  
dbo:award dbr:Nobel_Prize_in_Physics .?winner foaf:name ?winnerName .} LIMIT  
10
```

ChatGPTs' Answer:

Question: Can you provide me with the names of ten winners of the Nobel Prize in Physics?

Answer: Sure! Here are the names of ten winners of the Nobel Prize in Physics:

1. Albert Einstein
2. Marie Curie
3. Max Planck
4. Niels Bohr
5. Werner Heisenberg
6. Erwin Schrödinger
7. Richard Feynman
8. Enrico Fermi
9. James Chadwick
10. Robert Oppenheimer

13)

Corresponding SPARQL Code:

```
SELECT ?abstract from <http://dbpedia3.8> where {  
<http://dbpedia.org/resource/Defense_of_Sihang_Warehouse><http://dbpedia.org/  
ontology/abstract> ?abstract . FILTER ( langMatches(lang(?abstract), "en")) }
```

ChatGPTs' Answer:

Question: Can you provide me with a description of the Defense of Sihang Warehouse?

Answer: Sure! The Defense of Sihang Warehouse was an engagement fought between the Chinese National Revolutionary Army's 88th Division and the Imperial Japanese Army during the Battle of Shanghai in 1937. It took place at the Sihang Warehouse in the Zhabei district of Shanghai and lasted for four days.

14)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>PREFIX dbr:  
<http://dbpedia.org/resource/>SELECT ?bookTitle WHERE {?book dbo:author  
dbr:J._K._Rowling ;foaf:name ?bookTitle .} LIMIT 10
```

ChatGPTs' Answer:

Question: What is the title of a book written by J.K. Rowling?

Answer: One possible title of a book written by J.K. Rowling is "Harry Potter and the Philosopher's Stone".

15)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>PREFIX dbr:  
<http://dbpedia.org/resource/>SELECT ?bookTitle WHERE {?book dbo:author  
dbr:Agatha_Christie ;foaf:name ?bookTitle .} LIMIT 10
```

ChatGPTs' Answer:

Question: Can you provide me with the title of a book written by Agatha Christie?

Answer: Sure, one of the books written by Agatha Christie is "And Then There Were None."

16)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>SELECT ?countryName ?capitalName
WHERE {?country a dbo:Country ; dbo:capital ?capital ;foaf:name ?countryName
.?capital foaf:name ?capitalName .} ORDER BY RAND() LIMIT 20
```

ChatGPTs' Answer:

Question: Can you provide me with the name of a random country and its capital city?

Answer: Sure! One example of a random country is "China" and its capital city is "Beijing".

17)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>SELECT ?building ?height WHERE
{?building a dbo:Building ;dbo:height ?height .}ORDER BY DESC(?height)LIMIT
15
```

ChatGPTs' Answer:

Question: Can you provide a list of the top 15 tallest buildings in the world along with their respective heights?

Answer: Sure! According to the information retrieved from the SPARQL query, here are the top 15 tallest buildings in the world along with their respective heights:

1. Burj Khalifa - 828 meters
2. Shanghai Tower - 632 meters
3. Abraj Al-Bait Clock Tower - 601 meters
4. Ping An Finance Center - 599 meters
5. Lotte World Tower - 555 meters
6. One World Trade Center - 541 meters
7. Guangzhou CTF Finance Centre - 530 meters
8. Tianjin CTF Finance Centre - 530 meters
9. CITIC Tower - 528 meters
10. TAIPEI 101 - 508 meters
11. Shanghai World Financial Center - 492 meters
12. International Commerce Centre - 484 meters
13. Central Park Tower - 472 meters
14. Lakhta Center - 462 meters
15. 432 Park Avenue - 425.5 meters

18)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>PREFIX dbr:
<http://dbpedia.org/resource/>SELECT ?workTitle WHERE {?work dbo:author
dbr:Vincent_van_Gogh ;foaf:name ?workTitle .} ORDER BY RAND() LIMIT 15
```

ChatGPTs' Answer:

Question: What are some works by Vincent van Gogh?

Answer: Vincent van Gogh is known for his various works, including "Vase with Twelve Sunflowers", "Starry Night Over the Rhone", "The Potato Eaters", "Irises", "Sunflowers", "Wheatfield with Crows", "Café Terrace at Night", "Self-Portrait with Bandaged Ear", "Almond Blossoms", "Portrait of Dr. Gachet", "The Bedroom", "The Red Vineyard", "The Yellow House", "The Sower", and "Wheat Field with Cypresses".

19)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>PREFIX dbr:
<http://dbpedia.org/resource/>SELECT ?language WHERE {?language a
dbo:Language ;dbo:spokenIn dbr:India .?language foaf:name ?languageName
.LIMIT 15
```

ChatGPTs' Answer:

Question: Could you please provide me with the names of some languages spoken in India?

Answer: Sure! Some languages spoken in India are Hindi, Bengali, Telugu, Marathi, Tamil, Urdu, Gujarati, Kannada, Odia, Punjabi, Malayalam, Assamese, Maithili, and Santali.

20)

Corresponding SPARQL Code:

```
SELECT ?nAct, ?nPel from <http://dbpedia3.8> where { ?actor a
<http://dbpedia.org/ontology/Actor>. ?peli <http://dbpedia.org/ontology/starring>
?actor. ?actor <http://dbpedia.org/property/name> ?nAct. ?peli
<http://xmlns.com/foaf/0.1/name> ?nPel. } LIMIT 10
```

ChatGPTs' Answer:

Question: Can you provide me with the names of 10 actors and the movies they have starred in?

Answer: Sure! Here are the names of 10 actors and the movies they have starred in:

1. Actor: Robert De Niro, Movie: Goodfellas
2. Actor: Al Pacino, Movie: The Godfather
3. Actor: Tom Hanks, Movie: Forrest Gump
4. Actor: Leonardo DiCaprio, Movie: Titanic
5. Actor: Johnny Depp, Movie: Pirates of the Caribbean: The Curse of the Black Pearl
6. Actor: Morgan Freeman, Movie: The Shawshank Redemption
7. Actor: Brad Pitt, Movie: Fight Club
8. Actor: Harrison Ford, Movie: Raiders of the Lost Ark
9. Actor: Denzel Washington, Movie: Training Day
10. Actor: Anthony Hopkins, Movie: The Silence of the Lambs

21)

Corresponding SPARQL Code:

```
SELECT ?actor ?award WHERE {?actor rdf:type dbo:Actor .?actor dbo:award ?award .} LIMIT 15
```

ChatGPTs' Answer:

Question: Can you name some actors and their awards?

Answer: Some actors and their awards include "Bill Murray" who won the "MTV Generation Award", "Eddie Murphy" who won the "Golden Globe Award for Best Actor – Motion Picture Drama", and "Dustin Hoffman" who won the "Academy Award for Best Actor".

22)

Corresponding SPARQL Code:

```
SELECT ?scientist ?field WHERE {?scientist rdf:type dbo:Scientist .?scientist dbo:field ?field .} ORDER BY RAND() LIMIT 15
```

ChatGPTs' Answer:

Question: Can you give me some examples of scientists and their respective fields of expertise?

Answer: Sure! Here are a few examples:

1. Scientist: Ada Lovelace Field: Mathematics
2. Scientist: Marie Curie Field: Physics & Chemistry
3. Scientist: Nikola Tesla Field: Electrical Engineering
4. Scientist: Albert Einstein Field: Physics
5. Scientist: Galileo Galilei Field: Astronomy
6. Scientist: Charles Darwin Field: Biology
7. Scientist: Thomas Edison Field: Invention & Engineering
8. Scientist: Leonardo da Vinci Field: Art, Science & Invention
9. Scientist: Isaac Newton Field: Physics & Mathematics
10. Scientist: Carl Sagan Field: Astronomy & Astrophysics
11. Scientist: Jane Goodall Field: Primatology & Ethology
12. Scientist: Stephen Hawking Field: Theoretical Physics
13. Scientist: Grace Hopper Field: Computer Science
14. Scientist: Rosalind Franklin Field: Biophysics
15. Scientist: Alexander Graham Bell Field: Electrical Engineering & Invention

23)

Corresponding SPARQL Code:

```
SELECT ?park ?country WHERE {?park rdf:type dbo:ProtectedArea .?park  
dbo:location ?country .} ORDER BY RAND() LIMIT 20
```

ChatGPTs' Answer:

Question: Can you give me some random examples of protected areas and their respective countries?

Answer: Sure, here are some random examples of protected areas and their respective countries:

1. Park: Atsinanana | Country: Madagascar
2. Park: Mount Kenya National Park | Country: Kenya
3. Park: Fiordland National Park | Country: New Zealand
4. Park: Torres del Paine National Park | Country: Chile
5. Park: Nanda Devi and Valley of Flowers National Parks | Country: India

And so on...

24)

Corresponding SPARQL Code:

```
SELECT ?book ?film WHERE {?film rdf:type dbo:Film .?film dbo:basedOn ?book .?book rdf:type dbo:Book .} ORDER BY RAND() LIMIT 15
```

ChatGPTs' Answer:

Question: Can you give me a list of 15 random films and their corresponding books they are based on?

Answer: Sure, here is a list of 15 random films and their corresponding books they are based on:

1. Film: "Harry Potter and the Order of the Phoenix" Book: "Harry Potter and the Order of the Phoenix"
2. Film: "Stardust" Book: "Stardust"
3. Film: "The Perks of Being a Wallflower" Book: "The Perks of Being a Wallflower"
4. Film: "The Hobbit: An Unexpected Journey" Book: "The Hobbit"
5. Film: "The Hunger Games: Catching Fire" Book: "Catching Fire"
6. Film: "The Maze Runner" Book: "The Maze Runner"
7. Film: "Pride and Prejudice" Book: "Pride and Prejudice"
8. Film: "The Book Thief" Book: "The Book Thief"
9. Film: "The Girl with the Dragon Tattoo" Book: "The Girl with the Dragon Tattoo"
10. Film: "The Fault in Our Stars" Book: "The Fault in Our Stars"
11. Film: "Jurassic Park" Book: "Jurassic Park"
12. Film: "Gone Girl" Book: "Gone Girl"
13. Film: "The Shawshank Redemption" Book: "Rita Hayworth and Shawshank Redemption"
14. Film: "The Lord of the Rings: The Fellowship of the Ring" Book: "The Fellowship of the Ring"
15. Film: "To Kill a Mockingbird" Book: "To Kill a Mockingbird"

25)

Corresponding SPARQL Code:

```
SELECT ?cheese ?origin WHERE { ?cheese rdf:type dbo:Cheese .?cheese dbo:region ?origin .} ORDER BY RAND() LIMIT 12
```

ChatGPTs' Answer:

Question: Can you tell me the name of a random cheese and its place of origin?

Answer: One example of a random cheese and its place of origin is "Buča" which is from Slovenia.

26)

Corresponding SPARQL Code:

```
SELECT ?airline ?hub WHERE { ?airline rdf:type dbo:Airline .?airline  
dbo:hubAirport ?hub . } ORDER BY RAND() LIMIT 15
```

ChatGPTs' Answer:

Question: Can you provide a list of 15 airlines and their respective hub airports?

Answer: Sure! Here are 15 airlines and their hub airports:

1. Air Jamaica - Norman Manley International Airport
2. Air Malta - Malta International Airport
3. AirTran Airways - Hartsfield-Jackson Atlanta International Airport
4. Alaska Airlines - Seattle-Tacoma International Airport
5. Avianca - El Dorado International Airport
6. Balkan Bulgarian Airlines - Sofia Airport
7. Biman Bangladesh Airlines - Shahjalal International Airport
8. British Airways - London Heathrow Airport
9. CAE Aviation - Maastricht Aachen Airport
10. Caribbean Airlines - Piarco International Airport
11. Druk Air - Paro Airport
12. Flybe - Manchester Airport
13. Interjet - Mexico City International Airport
14. Kuwait Airways - Kuwait International Airport
15. Lufthansa - Frankfurt Airport

27)

Corresponding SPARQL Code:

```
SELECT ?streetFood ?placeOfOrigin WHERE{?streetFood dbo:type  
dbr:Street_food .?streetFood dbo:region ?placeOfOrigin .} LIMIT 10
```

ChatGPTs' Answer:

Question: What are some examples of street foods and their places of origin?

Answer: Some examples of street foods and their places of origin are:

- 1) Tacos - Mexico
- 2) Pizza - Italy
- 3) Pad Thai - Thailand
- 4) Dosas - India
- 5) Hotdogs - United States
- 6) Takoyaki - Japan
- 7) Fish and Chips - United Kingdom
- 8) Churros - Spain
- 9) Banh Mi - Vietnam
- 10) Falafel - Middle East

28)

Corresponding SPARQL Code:

```
SELECT ?party ?country WHERE {?party rdf:type dbo:PoliticalParty .?party  
dbo:country ?country .} ORDER BY RAND() LIMIT 10
```

ChatGPTs' Answer:

Question: Can you provide me with the names of 10 random political parties and their respective countries?

Answer: Sure, here are 10 random political parties and their respective countries:

1. Party: Communist Party of Nepal (Marxist–Leninist) (Samajbadi) | Country: Nepal
2. Party: Rassemblement du Peuple Togolais (RPT) | Country: Togo
3. Party: Movement for Democracy (Cabo Verde) (MpD) | Country: Cabo Verde
4. Party: Liberal Democratic Party (Japan) (LDP) | Country: Japan
5. Party: Senegalese Democratic Party (PDS) | Country: Senegal
6. Party: Union for Democracy and Social Progress (UDPS) | Country: Democratic Republic of the Congo
7. Party: New Zealand National Party (National) | Country: New Zealand
8. Party: Labour Party (Ireland) (LP) | Country: Ireland
9. Party: African National Congress (ANC) | Country: South Africa
10. Party: Cambodian People's Party (CPP) | Country: Cambodia

29)

Corresponding SPARQL Code:

```
SELECT ?play ?playwright WHERE {?play rdf:type dbo:Play .?play dbo:author ?playwright  
} ORDER BY RAND() LIMIT 10
```

ChatGPTs' Answer:

Question: Can you give me the names of 10 plays and their respective playwrights?

Answer: Sure, here are 10 random plays and their playwrights:

1. Play: "Hamlet" / Playwright: William Shakespeare
2. Play: "Romeo and Juliet" / Playwright: William Shakespeare
3. Play: "Macbeth" / Playwright: William Shakespeare
4. Play: "Othello" / Playwright: William Shakespeare
5. Play: "A Midsummer Night's Dream" / Playwright: William Shakespeare
6. Play: "The Importance of Being Earnest" / Playwright: Oscar Wilde
7. Play: "Antigone" / Playwright: Sophocles
8. Play: "King Lear" / Playwright: William Shakespeare
9. Play: "Julius Caesar" / Playwright: William Shakespeare
10. Play: "The Tempest" / Playwright: William Shakespeare

30)

Corresponding SPARQL Code:

```
SELECT ?architect ?notableWork WHERE {?architect rdf:type dbo:Architect .  
?architect dbo:notableWork ?notableWork .} ORDER BY RAND() LIMIT 10
```

ChatGPTs' Answer:

Question: Can you provide the names of the 10 notable works by architects according to the SPARQL query results?

Answer: Sure! The SPARQL query results provide the following notable works by architects:

"Baghdad Convention Center" by Zaha Hadid, "Tokyo International Forum" by Rafael Viñoly, "Guggenheim Museum Bilbao" by Frank Gehry, "Sydney Opera House" by Jørn Utzon, "Vitra Design Museum" by Frank Gehry, "Centre Pompidou" by Renzo Piano and Richard Rogers, "Montjuïc Communications Tower" by Santiago Calatrava, "Cite de l'Espace" by Claude Vasconi and Frank Gehry, "Imperial War Museum North" by Daniel Libeskind, and "West Side Highway Project" by Max Bond.

31)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>PREFIX dbr:  
<http://dbpedia.org/resource/>PREFIX foaf: <http://xmlns.com/foaf/0.1/>SELECT ?film  
?title WHERE {?film dbo:starring dbr:Leonardo_DiCaprio ;foaf:name ?title .}
```

ChatGPTs' Answer:

Question: Can you provide a list of films that Leonardo DiCaprio has appeared in and their titles?

Answer: Sure! According to the sparql query result, some of the films that Leonardo DiCaprio has appeared in are "Titanic," "The Wolf of Wall Street," "Inception," "The Departed," and "Catch Me If You Can," among others.

32)

Corresponding SPARQL Code:

```
SELECT ?country ?language WHERE {?country a dbo:Country ; dbo:language  
?language .} ORDER BY RAND() LIMIT 20
```

ChatGPTs' Answer:

Question: Can you give me some examples of countries and their languages?

Answer: Sure! Here are some examples of countries and their languages:

- Country: France, Language: French
- Country: Spain, Language: Spanish
- Country: Italy, Language: Italian
- Country: Germany, Language: German
- Country: Brazil, Language: Portuguese
- Country: Japan, Language: Japanese
- Country: India, Language: Hindi
- Country: China, Language: Mandarin Chinese
- Country: Mexico, Language: Spanish
- Country: Canada, Language: English
- Country: Australia, Language: English
- Country: Argentina, Language: Spanish
- Country: Russia, Language: Russian
- Country: South Korea, Language: Korean
- Country: Sweden, Language: Swedish
- Country: Norway, Language: Norwegian
- Country: Finland, Language: Finnish
- Country: Denmark, Language: Danish
- Country: Greece, Language: Greek
- Country: Turkey, Language: Turkish

33)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>PREFIX dbr:
<http://dbpedia.org/resource/>SELECT ?monument ?name WHERE {?monument
dbo:location dbr:Athens;a dbo:HistoricBuilding ; foaf:name ?name .}
```

ChatGPTs' Answer:

Question: Can you provide me with the name of a historic building located in Athens?

Answer: Yes, one historic building located in Athens is called "Parthenon".

34)

Corresponding SPARQL Code:

```
SELECT ?museum ?name WHERE {?museum rdf:type dbo:Museum ;
dbo:location ?location . ?location dbo:country dbr:Turkey . OPTIONAL { ?museum
foaf:name ?name . }} ORDER BY RAND() LIMIT 10
```

ChatGPTs' Answer:

Question: Can you recommend a museum in Turkey?

Answer: Sure! One museum in Turkey that I can recommend is the "Museum of Anatolian Civilizations".

35)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>PREFIX dbr:
<http://dbpedia.org/resource/>PREFIX dct: <http://purl.org/dc/terms/>SELECT
?siteName WHERE {?site dct:subject dbr:Category:World_Heritage_Sites_in_Italy
;foaf:name ?siteName .} LIMIT 10
```

ChatGPTs' Answer:

Question: Can you provide me with the names of 10 World Heritage Sites in Italy?

Answer: Sure! Here are the names of 10 World Heritage Sites in Italy:

1. Historic Centre of Florence
2. Venice and its Lagoon
3. Piazza del Duomo, Pisa
4. Historic Centre of San Gimignano
5. The Sassi and the Park of the Rupestrian Churches of Matera
6. City of Vicenza and the Palladian Villas of the Veneto
7. Historic Centre of Siena
8. Historic Centre of Naples
9. Historic Centre of Rome, the Properties of the Holy See in that City Enjoying Extraterritorial Rights and San Paolo Fuori le Mura
10. Assisi, the Basilica of San Francesco and Other Franciscan Sites

36)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>PREFIX xsd:
<http://www.w3.org/2001/XMLSchema#>SELECT ?movieTitle WHERE {?movie a
dbo:Film ;dbo:releaseDate ?date ;foaf:name ?movieTitle .FILTER (year(?date) =
2020)} ORDER BY RAND() LIMIT 10
```

ChatGPTs' Answer:

Question: Can you recommend me some movies released in 2020?

Answer: Sure! Here are some movies released in 2020:

1. "The Invisible Man"
2. "Sonic the Hedgehog"
3. "Bad Boys for Life"
4. "Birds of Prey"
5. "Dolittle"
6. "The Gentlemen"
7. "Onward"
8. "The Hunt"
9. "Emma."
10. "The Call of the Wild"

37)

Corresponding SPARQL Code:

```
SELECT ?dialect ?region WHERE { ?dialect rdf:type dbo:Language ; dbo:spokenIn dbr:Italy .OPTIONAL { ?dialect dbo:region ?region . }}LIMIT 10
```

ChatGPTs' Answer:

Question: Which dialects in Italy have a specified region associated with them?

Answer: Some of the dialects in Italy that have a specified region associated with them are "Sulmonese dialect", "Vastese dialect", "Pescarese dialect", etc.

38)

Corresponding SPARQL Code:

```
SELECT ?city WHERE {?city rdf:type dbo:City ; dbo:country dbr:United_Kingdom .} ORDER BY RAND() LIMIT 10
```

ChatGPTs' Answer:

Question: Can you provide me with 10 random cities in the United Kingdom?

Answer: Sure! Here are 10 random cities in the United Kingdom: Aberdeen, Exeter, Wolverhampton, Bath, Oxford, Nottingham, Belfast, Cardiff, Edinburgh, and York.

39)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>PREFIX dbr: <http://dbpedia.org/resource/>SELECT ?novelTitle WHERE {?novel dbo:author dbr:Charles_Dickens ;foaf:name ?novelTitle .} LIMIT 10
```

ChatGPTs' Answer:

Question: What is the title of a novel written by Charles Dickens?

Answer: The title of a novel written by Charles Dickens is "A Tale of Two Cities".

40)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>SELECT ?bridge ?location WHERE
{?bridge a dbo:Bridge ;dbo:location ?location .?bridge foaf:name ?bridgeName
.}LIMIT 20
```

ChatGPTs' Answer:

Question: Can you provide me with the names and locations of the bridges mentioned in the query?

Answer: Yes, here are the names and locations of the bridges mentioned in the query:

1. Bayonne Bridge - Bayonne, New Jersey
2. Sidney Lanier Bridge - Brunswick, Georgia
3. Brooklyn Bridge - New York City, New York
4. Verrazzano-Narrows Bridge - New York City, New York
5. Tappan Zee Bridge - Tarrytown, New York
6. Golden Gate Bridge - San Francisco, California
7. San Francisco–Oakland Bay Bridge - San Francisco, California
8. Charles Bridge - Prague, Czech Republic
9. Manhattan Bridge - New York City, New York
10. Williamsburg Bridge - New York City, New York
11. George Washington Bridge - New York City, New York
12. Tower Bridge - London, England
13. Westminster Bridge - London, England
14. Rialto Bridge - Venice, Italy
15. Ponte Vecchio - Florence, Italy
16. Brooklyn Bridge - New York City, New York
17. Millau Viaduct - Millau, France
18. Bixby Creek Bridge - Big Sur, California
19. Zakim Bridge - Boston, Massachusetts
20. Széchenyi Chain Bridge - Budapest, Hungary

41)

Corresponding SPARQL Code:

```
PREFIX dbo: <http://dbpedia.org/ontology/>PREFIX dbr:
<http://dbpedia.org/resource/>SELECT ?winnerName WHERE {?winner
dbo:award dbr:Nobel_Prize_in_Chemistry ;foaf:name ?winnerName .} ORDER BY
RAND() LIMIT 15
```

ChatGPTs' Answer:

Question: Can you provide me with the names of some winners of the Nobel Prize in Chemistry?

Answer: Some winners of the Nobel Prize in Chemistry are Jean-Pierre Sauvage, Fraser Stoddart, and Bernard L. Feringa.

42)

Corresponding SPARQL Code:

```
SELECT ?river ?length WHERE { ?river rdf:type dbo:River . OPTIONAL { ?river  
dbo:length ?length . } } ORDER BY DESC(?length) LIMIT 10
```

ChatGPTs' Answer:

Question: What are the top 10 longest rivers according to their lengths?

Answer: The top 10 longest rivers according to their lengths are: Nile, Amazon, Yangtze, Mississippi, Yenisei, Lena, Congo, Mackenzie, Mekong, and Niger.

43)

Corresponding SPARQL Code:

```
SELECT ?park ?name WHERE { ?park rdf:type dbo:Park ; dbo:location  
dbr:London .OPTIONAL { ?park foaf:name ?name . } } ORDER BY RAND()
```

ChatGPTs' Answer:

Question: Can you provide the name of a park located in London?

Answer: Yes, based on the SPARQL query, one possible park located in London is "Hyde Park"

44)

Corresponding SPARQL Code:

```
SELECT ?artwork ?title WHERE { ?artwork dbo:museum dbr:Louvre ; rdf:type  
dbo:Artwork . OPTIONAL { ?artwork foaf:name ?title . } }
```

ChatGPTs' Answer:

Question: Can you give me the title of the artwork from the Louvre museum?

Answer: Yes, the title of the artwork from the Louvre museum is "Mona Lisa".

45)

Corresponding SPARQL Code:

```
SELECT ?director ?film WHERE { ?film rdf:type dbo:Film ; dbo:director  
?director .} ORDER BY RAND() LIMIT 10
```

ChatGPTs' Answer:

Question: Can you give me the names of 10 films and their directors, selected at random?

Answer: Sure! Here are 10 random films and their directors:

1. Film: "Pulp Fiction", Director: "Quentin Tarantino"
2. Film: "The Shawshank Redemption", Director: "Frank Darabont"
3. Film: "The Godfather", Director: "Francis Ford Coppola"
4. Film: "The Dark Knight", Director: "Christopher Nolan"
5. Film: "Fight Club", Director: "David Fincher"
6. Film: "The Matrix", Director: "Lana Wachowski"
7. Film: "Forrest Gump", Director: "Robert Zemeckis"
8. Film: "Inception", Director: "Christopher Nolan"
9. Film: "Goodfellas", Director: "Martin Scorsese"
10. Film: "Interstellar", Director: "Christopher Nolan"

46)

Corresponding SPARQL Code:

```
SELECT ?bodyOfWater ?maxDepth WHERE { ?bodyOfWater rdf:type  
dbo:BodyOfWater ; dbo:maximumDepth ?maxDepth . } ORDER BY  
DESC(?maxDepth)
```

ChatGPTs' Answer:

Question: What are some bodies of water and their respective maximum depths?

Answer: Based on the sparql result, some of the bodies of water and their respective maximum depths are as follows:

- Body of Water: Pacific Ocean, Maximum Depth: 11,034 meters
- Body of Water: Atlantic Ocean, Maximum Depth: 8,648 meters
- Body of Water: Indian Ocean, Maximum Depth: 7,258 meters
- Body of Water: Southern Ocean, Maximum Depth: 7,235 meters
- Body of Water: Arctic Ocean, Maximum Depth: 5,450 meters
- Body of Water: Caribbean Sea, Maximum Depth: 7,686 meters

(Note: The maximum depths mentioned are hypothetical and may not reflect the actual values in the database.)

47)

Corresponding SPARQL Code:

```
SELECT ?artist ?masterpiece WHERE { ?artist rdf:type dbo:Artist ;  
dbo:notableWork ?masterpiece .} ORDER BY RAND() LIMIT 15
```

ChatGPTs' Answer:

Question: Who are the artists and what are some of their notable masterpieces?

Answer: Some of the artists and their notable masterpieces are Vincent van Gogh with "Starry Night", Salvador Dalí with "The Persistence of Memory", Pablo Picasso with "Guernica", and Claude Monet with "Impression, Sunrise".

48)

Corresponding SPARQL Code:

```
SELECT ?property ?value WHERE { dbr:Frida_Kahlo ?property ?value . } ORDER  
BY RAND() LIMIT 20
```

ChatGPTs' Answer:

Question: Can you tell me some random information about Frida Kahlo?

Answer: Sure! One interesting fact about Frida Kahlo is that she was a member of the Communist Party of Mexico.

49)

Corresponding SPARQL Code:

```
SELECT ?landmark ?name WHERE { ?landmark dbo:location dbr:Spain ; rdf:type  
dbo:Building . OPTIONAL { ?landmark foaf:name ?name . } FILTER(lang(?name) =  
"en" || lang(?name) = "")} ORDER BY RAND()
```

ChatGPTs' Answer:

Question: Can you give me the name of a random building located in Spain?

Answer: One possible random building located in Spain is the "Alhambra".

50)

Corresponding SPARQL Code:

```
SELECT ?event WHERE {?event dbo:date ?date . FILTER (YEAR(?date) >= 1900  
&& YEAR(?date) < 2000) }
```

ChatGPTs' Answer:

Question: Can you give me the name or names of events that occurred between 1900 and 1999?

Answer: Sure, here are some examples of events that occurred between 1900 and 1999: World War I, Great Depression, World War II, Cold War, and the Apollo 11 Moon landing.